Bit Manipulation Study Guide

1. Introduction to Bit Manipulation

Bit manipulation involves working with individual bits in binary representations of numbers. In Java, integers are represented using 32 bits, and bit manipulation allows us to perform operations directly on these bits for efficient programming.

Binary Number System

- Binary uses only 0s and 1s
- Each position represents a power of 2
- Example: 5 in binary is 101 ($1\times2^2 + 0\times2^1 + 1\times2^0$)

Why Use Bit Manipulation?

- Faster execution than arithmetic operations
- Memory efficient
- Useful in competitive programming
- Essential for system-level programming

2. Bitwise Operators in Java

AND Operator (&)

Returns 1 if both bits are 1, otherwise 0.

```java
int a = 5;    // 101 in binary
int b = 3;    // 011 in binary
int result = a & b;  // 001 in binary = 1
System.out.println(result); // Output: 1
```

OR Operator (|)

Returns 1 if at least one bit is 1.

```java
int a = 5;    // 101 in binary
int b = 3;    // 011 in binary
int result = a | b;  // 111 in binary = 7
System.out.println(result); // Output: 7
```

XOR Operator (^)

Returns 1 if bits are different, 0 if same.

```java
int a = 5;    // 101 in binary
int b = 3;    // 011 in binary
int result = a ^ b;  // 110 in binary = 6
System.out.println(result); // Output: 6
```

NOT Operator (~)

Flips all bits (bitwise complement).

```java
int a = 5;    // 101 in binary
int result = ~a;  // Flips all 32 bits
System.out.println(result); // Output: -6
```

Left Shift Operator (<<)

Shifts bits to the left by specified positions.

```java
int a = 5;    // 101 in binary
int result = a << 2;  // 10100 in binary = 20
System.out.println(result); // Output: 20
```

Right Shift Operator (>>)

Shifts bits to the right by specified positions.

```java
int a = 20;   // 10100 in binary
int result = a >> 2;  // 101 in binary = 5
System.out.println(result); // Output: 5
```

Unsigned Right Shift Operator (>>>)

Shifts bits to the right, filling with zeros.

```java
int a = -8;
int result = a >>> 2;
System.out.println(result); // Fills with 0s from left
```

3. Common Bit Manipulation Techniques

Check if a Number is Even or Odd

```java
public static boolean isEven(int n) {
    return (n & 1) == 0;
}

// Usage
System.out.println(isEven(4)); // true
System.out.println(isEven(7)); // false
```

Check if nth Bit is Set

```java
public static boolean isBitSet(int num, int n) {
    return (num & (1 << n)) != 0;
}

// Usage
System.out.println(isBitSet(5, 0)); // true (5 = 101, bit 0 is set)
System.out.println(isBitSet(5, 1)); // false (5 = 101, bit 1 is not set)
```

Set the nth Bit

```java
public static int setBit(int num, int n) {
    return num | (1 << n);
}

// Usage
System.out.println(setBit(5, 1)); // 7 (101 becomes 111)
```

Clear the nth Bit

```java
public static int clearBit(int num, int n) {
    return num & ~(1 << n);
}

// Usage
System.out.println(clearBit(5, 2)); // 1 (101 becomes 001)
```

Toggle the nth Bit

```java
public static int toggleBit(int num, int n) {
    return num ^ (1 << n);
}

// Usage
System.out.println(toggleBit(5, 1)); // 7 (101 becomes 111)
```

Count Number of Set Bits

```java
public static int countSetBits(int n) {
    int count = 0;
    while (n != 0) {
        count++;
        n = n & (n - 1); // Remove rightmost set bit
    }
    return count;
}

// Usage
System.out.println(countSetBits(7)); // 3 (111 has 3 set bits)
```

Find Power of 2

```java
public static boolean isPowerOfTwo(int n) {
    return n > 0 && (n & (n - 1)) == 0;
}

// Usage
System.out.println(isPowerOfTwo(8)); // true
System.out.println(isPowerOfTwo(6)); // false
```

Swap Two Numbers without Extra Variable

```java
public static void swap(int a, int b) {
    System.out.println("Before: a = " + a + ", b = " + b);
    a = a ^ b;
    b = a ^ b;
    a = a ^ b;
    System.out.println("After: a = " + a + ", b = " + b);
}

// Usage
swap(5, 3);
```

4. Advanced Bit Manipulation Techniques

Find the Only Non-Duplicate Number

```java
public static int findSingle(int[] arr) {
    int result = 0;
    for (int num : arr) {
        result ^= num;
    }
    return result;
}

// Usage: [2, 3, 2, 4, 3] returns 4
```

Find Missing Number in Array

```java
public static int findMissing(int[] arr, int n) {
    int xor1 = 0; // XOR of all array elements
    int xor2 = 0; // XOR of all numbers from 1 to n+1

    for (int i = 0; i < n; i++) {
        xor1 ^= arr[i];
    }

    for (int i = 1; i <= n + 1; i++) {
        xor2 ^= i;
    }

    return xor1 ^ xor2;
}
```

Generate All Subsets using Bit Manipulation

```java
public static void generateSubsets(int[] arr) {
    int n = arr.length;
    for (int i = 0; i < (1 << n); i++) {
        System.out.print("{ ");
        for (int j = 0; j < n; j++) {
            if ((i & (1 << j)) != 0) {
                System.out.print(arr[j] + " ");
            }
        }
        System.out.println("}");
    }
}
```

5. Practice Questions on Bit Manipulation

Basic Level Problems

1. Write a program to reverse the bits of a given integer
2. Find the position of the rightmost set bit
3. Check if a number has alternating bits (101010...)
4. Count total set bits in all numbers from 1 to N

Intermediate Level Problems

1. Find two non-repeating elements in an array where every other element repeats twice
2. Multiply a number by 7 without using multiplication operator
3. Find the element that appears once in a sorted array where all other elements appear twice
4. Check if binary representation of a number is palindrome

Advanced Level Problems

1. Given an array where every element occurs three times, except one element which occurs only once, find that element
2. Find the maximum XOR of any two elements in an array
3. Count number of bits to be flipped to convert A to B
4. Find the longest sequence of 1s that can be achieved by flipping at most one bit

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Quiz Questions

Question 1: What is the result of 5 & 3 in Java? a) 1 b) 7 c) 6 d) 0

Answer: a) 1

Question 2: Which operator is used to set the nth bit of a number? a) & b) | c) ^ d) ~

Answer: b) |

Question 3: What does the expression (n & (n-1)) == 0 check for? a) If n is even b) If n is odd c) If n is a power of 2 d) If n is prime

Answer: c) If n is a power of 2

Question 4: What is the result of 8 >> 2? a) 32 b) 4 c) 2 d) 16

Answer: c) 2

Question 5: Which operation can be used to swap two numbers without using a temporary variable? a) AND b) OR c) XOR d) NOT

Answer: c) XOR

Practice Coding Questions

Question 1: Write a Java method that takes an integer and returns the number of 1s in its binary representation.

Question 2: Given an array of integers where every element appears twice except for one, write a method to find that single element using bit manipulation.

Question 3: Write a method that checks whether the binary representation of a given positive integer is a palindrome.

Question 4: Given two integers, write a method that returns the number of bit positions where the two numbers have different bits.

Question 5: Write a method that finds the next power of 2 greater than or equal to a given number using bit manipulation techniques.